



DATA ANALYTICS PROJECT PORTFOLIO

Salaries of Jobs in the Data Analyst Domain By Humsin Faisal Akbar

Implementing SQL for a Business Case Study

1. Case Study Background

A company wants to conduct a comprehensive analysis of salary patterns for Data Analyst employees in their industry, as well as understand how factors such as experience level, job type, and company characteristics affect employee salaries.

To address this need, data exploration was carried out using SQL on a salary dataset for jobs in the data science/analytics field.

2. Dataset Information

The dataset contains salary data for employees across various data-related positions, with the following columns:

Column	Description
work_year	The year the salary was paid.
experience_level	Experience level: EN (Entry-level/Junior), MI (Mid-level/Intermediate), SE (Senior-level/Expert), EX (Executive-level/Director).
employment_type	Type of employment: PT (Part-time), FT (Full-time), CT (Contract), FL (Freelance).
job_title	The role worked in during that year.
salary	The total gross salary amount paid.
salary_currency	The currency of the salary, as an ISO 4217 code.
salary_in_usd	The salary converted to USD (using the average exchange rate for the respective year).
employee_residence	Employee's primary country of residence (ISO 3166 code).
remote_ratio	Percentage of remote work: 0 (no remote work), 50 (partially remote), 100 (fully remote).
company_location	Country of the employer's main office or contracting branch (ISO 3166 code).
company_size	Company size: S (<50 employees), M (50–250 employees), L (>250 employees).

3. Analysis Process and SQL Queries

Below is a series of business questions answered using SQL, complete with the query, a result preview, and the resulting insight.

Question 1

Which job titles are related to the Data Analyst position?

Skill set: *Basic SQL, Filtering*

SQL Query:

```
SELECT
  DISTINCT job_title
FROM `sql-project-424814.ds_salaries.salaries`
WHERE
  job_title LIKE '%Data Analyst%';
```

Result Preview:

job_title
Data Analyst
BI Data Analyst
Business Data Analyst
Financial Data Analyst
Product Data Analyst
Principal Data Analyst
Lead Data Analyst
Finance Data Analyst
Marketing Data Analyst

Insight: There are 9 job titles related to Data Analyst in the dataset, reflecting the variety of specializations within this role.

Question 2

What is the average salary for data analysis jobs?

Skill set: *Filtering, Aggregation, Group By*

SQL Query:

```
SELECT
  DISTINCT job_title,
  AVG(salary_in_usd) as avg_salary_in_usd
FROM `sql-project-424814.ds_salaries.salaries`
WHERE
  job_title LIKE '%Data Analyst%'
GROUP BY 1 ;
```

Result Preview:

job_title	avg_salary_in_usd
Data Analyst	92,893.06
BI Data Analyst	74,755.17
Business Data Analyst	76,691.20
Financial Data Analyst	275,000.00
Product Data Analyst	13,036.00
Principal Data Analyst	122,500.00
Lead Data Analyst	92,203.00
Finance Data Analyst	61,896.00
Marketing Data Analyst	88,654.00

Insight: The average salary for data analysis jobs varies widely, ranging from about \$1.3k to \$275k, with Financial Data Analyst showing the highest average salary.

Question 3

What is the average salary for data analysis jobs based on experience level and employment type?

Skill set: *Filtering, Aggregation, Group By*

SQL Query:

```
SELECT
  job_title,
  experience_level,
  employment_type,
  AVG(salary_in_usd) AS avg_salary
FROM
  `sql-project-424814.ds_salaries.salaries`
WHERE
  job_title LIKE '%Data Analyst%'
GROUP BY
  job_title,
  experience_level,
  employment_type
ORDER BY
  avg_salary DESC;
```

Result Preview:

job_title	experience_level	employment_type	avg_salary
Financial Data Analyst	MI	FT	450,000.00
Lead Data Analyst	SE	FT	170,000.00
Principal Data Analyst	SE	FT	170,000.00
BI Data Analyst	EX	FT	150,000.00
Data Analyst	EX	FT	120,000.00
Data Analyst	SE	FT	111,922.63

Insight: The highest salary is found for Financial Data Analyst at the Mid-level with Full-time status, showing that the combination of job title, experience level, and employment type strongly affects salary.

Question 4

Which country offers a salary level above the average for full-time Data Analyst positions with entry or mid-level experience?

Skill set: Filtering, Aggregation, Group By, Subquery

SQL Query:

```
SELECT
  company_location,
  experience_level,
  employment_type,
  AVG(salary_in_usd) AS avg_salary
FROM
  `sql-project-424814.ds_salaries.salaries`
WHERE
  job_title LIKE '%Data Analyst%'
  AND experience_level IN ('EN','MI')
  AND employment_type = 'FT'
GROUP BY
  company_location,
  experience_level,
  employment_type
HAVING avg_salary > (
  SELECT
    AVG(salary_in_usd)
  FROM `sql-project-424814.ds_salaries.salaries`
  WHERE
    job_title LIKE '%Data Analyst%'
    AND experience_level IN ('EN','MI')
    AND employment_type = 'FT'
);
```

Result Preview:

company_location	experience_level	employment_type	avg_salary
US	EN	FT	74,750.00
US	MI	FT	111,548.52
CA	MI	FT	76,478.00

Insight: The United States (for Entry & Mid level) and Canada (for Mid level) offer salaries above the overall average, indicating that both countries are competitive for full-time Data Analyst positions.

Question 5

Which country has the highest and lowest average salary for entry-level 'Data Analyst' related positions with full-time employment?

Skill set: Filtering, Aggregation, Group By, Subquery, Limit, Set Operation

SQL Query:

```
(
  SELECT
    company_location,
    experience_level,
    employment_type,
    AVG(salary_in_usd) AS avg_salary
  FROM
    `sql-project-424814.ds_salaries.salaries`
  WHERE
    job_title LIKE '%Data Analyst%'
    AND experience_level IN ('EN')
    AND employment_type = 'FT'
  GROUP BY
    company_location, experience_level, employment_type
  ORDER BY avg_salary DESC
  LIMIT 1
)
UNION ALL
(
  SELECT
    company_location,
    experience_level,
    employment_type,
    AVG(salary_in_usd) AS avg_salary
  FROM
    `sql-project-424814.ds_salaries.salaries`
  WHERE
    job_title LIKE '%Data Analyst%'
    AND experience_level IN ('EN')
    AND employment_type = 'FT'
  GROUP BY
    company_location, experience_level, employment_type
  ORDER BY avg_salary ASC
  LIMIT 1
);
```

Result Preview:

company_location	experience_level	employment_type	avg_salary
US	EN	FT	74,750.00
IN	EN	FT	6,072.00

Insight: The United States has the highest average salary for entry-level, full-time Data Analyst positions, while India has the lowest average salary in the same category — indicating a significant salary gap between countries.

Question 6

In which year did the salary increase from entry to mid level have the highest increase for full-time Data Analyst-related jobs?

Skill set: Filtering, Aggregation, Group By, CTE, Joins

SQL Query:

```
WITH ds_1 AS (  
    SELECT  
        work_year,  
        AVG(salary_in_usd) AS avg_salary_en  
    FROM  
        `sql-project-424814.ds_salaries.salaries`  
    WHERE  
        job_title LIKE '%Data Analyst%'  
        AND experience_level = 'EN'  
        AND employment_type = 'FT'  
    GROUP BY  
        work_year  
)  
,  
ds_2 AS (  
    SELECT  
        work_year,  
        AVG(salary_in_usd) AS avg_salary_mi  
    FROM  
        `sql-project-424814.ds_salaries.salaries`  
    WHERE  
        job_title LIKE '%Data Analyst%'  
        AND experience_level = 'MI'  
        AND employment_type = 'FT'  
    GROUP BY  
        work_year  
)  
SELECT  
    ds_1.work_year,  
    ds_1.avg_salary_en,  
    ds_2.avg_salary_mi,  
    SUM(ds_2.avg_salary_mi - ds_1.avg_salary_en) AS difference  
FROM ds_1  
LEFT JOIN ds_2 ON ds_1.work_year = ds_2.work_year  
GROUP BY  
    ds_1.work_year, ds_1.avg_salary_en, ds_2.avg_salary_mi  
ORDER BY ds_1.work_year;
```

Result Preview:

work_year	avg_salary_en	avg_salary_mi	difference
2020	44,768.00	60,728.87	15,960.87
2021	57,809.50	108,398.82	50,589.32
2022	73,000.00	68,970.59	-4,029.41

Insight: The highest salary increase from entry to mid level occurred in 2021. Meanwhile, 2022 actually shows a negative difference, indicating the need for further analysis of external factors such as economic conditions during that year.

4. Key Insights

- There are 9 job titles related to data analysis in the dataset.
- The average salary for data analysis jobs ranges from about \$1.3k to \$275k.
- The highest average salary for data analysis is for Financial Data Analyst with mid-level experience.
- The United States offers above-average salaries for entry-level and mid-level full-time employment, while Canada offers above-average salaries for mid-level full-time employment.
- The United States has the highest average salary for entry-level, full-time data analysis jobs, while India has the lowest average salary in the same category.
- The highest salary increase from entry to mid level for full-time data analysis jobs occurred in 2020–2021.
- The negative difference in 2022 could be due to several reasons such as economic factors or other external events — further analysis is needed to confirm the cause.



5. Conclusion

This project demonstrates an end-to-end application of SQL to answer real business questions — from basic filtering and aggregation, to the use of subqueries and CTEs, through to set operations and joins across derived tables. The case study confirms that salaries for Data Analyst positions are strongly influenced by a combination of job title, experience level, employment type, company location, and work year.

Skills demonstrated in this project: Basic SQL, Filtering, Aggregation, Group By, Subquery, Limit, Set Operation, Common Table Expressions (CTE), and Joins.



Thank You

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